

Ghana Events Marketplace — Functional Specification

Document purpose: Provide a clear, developer-friendly functional specification for building the Event Services Discovery Platform (starting with Accra). This covers scope, user roles, functional requirements, data model, API endpoints, UI screens, acceptance criteria, non-functional requirements, MVP roadmap, monetization, and next steps.

1. Project Overview

Name (working): Ghana Events Marketplace

Purpose: A discovery and booking platform for event-related services in Ghana. Users find vendors (venues, caterers, photographers, DJs, fashion designers, fabrics, favours, drinks, event planners, etc.) by location and category, view profiles, contact or book vendors, and manage event planning.

Primary goals (MVP): - Provide searchable, location-aware listings for event vendors in Accra. - Offer vendor profiles with media, descriptions, contact, tags, and availability. - Enable vendor onboarding (self-serve) and admin verification. - Support monetization: subscriptions, featured listings and lead fees.

Assumptions: - Initially focused on Accra, Ghana, with manual data import of your vendor lists. - Booking flow for venues/caterers can be optional in MVP; start with contact/lead capture and expand to bookings.

2. Personas & User Roles

Personas

- **Event Planner / Consumer** — Users searching for vendors to hire for weddings, birthdays, corporate events.
- **Vendor Owner** — Venues, caterers, photographers who want leads and bookings.
- **Admin / Platform Manager** — Manages listings, payments, disputes, and analytics.

Roles & Permissions

- **Guest:** browse & search listings, view limited contact information, read reviews.
- **Registered User:** save favorites, request contact, leave reviews, message vendors, create event projects (optional).
- **Vendor:** manage profile, upload images, set pricing, availability, view leads, accept payments, view analytics.
- **Admin:** manage users/vendors/content, approve vendor profiles, manage billing, view platform analytics.

3. High-level Features (MVP and Post-MVP)

MVP Features

1. **Search & Discovery**
2. Search by category, keyword, and location (neighbourhood or map).
3. Filters: price range, availability, ratings, tags (e.g., wedding, outdoor).
4. **Listing Profiles**
5. Name, address, category, short description, gallery, contact options, tags, basic pricing tier, reviews.
6. **Vendor Onboarding**
7. Sign-up, profile creation wizard, media upload, category selection, verification request.
8. **Lead Capture**
9. "Request Quote" or "Show Phone" button triggers a lead stored in vendor dashboard and optionally triggers email/SMS.
10. **Vendor Dashboard**
11. View leads, edit profile, subscription management, performance metrics (views, clicks, leads).
12. **Admin Panel**
13. Approve/decline vendors, manage listings, view payments, run queries.
14. **Monetization**
15. Subscriptions for vendors, featured listing purchases, per-lead billing option.
16. **Basic Analytics**
17. Views, clicks, leads per vendor, top search terms.

Post-MVP features

- Full booking flow with calendar sync and payments (commission model).
- In-app chat & scheduling.
- Reviews & disputes system.
- Packages / Bundled services.
- Multi-city rollout and geospatial clustering.
- API for partners and vendors.

4. User Stories & Acceptance Criteria (Representative)

User Story A — Search for a venue

As a user I want to search for venues in East Legon and see results ranked by relevance and proximity **So that** I can find a venue close to my location.

Acceptance criteria: - Search UI accepts location text or uses user location. - Results show name, thumbnail, distance, rating and a short snippet. - Clicking a result opens the vendor profile page.

User Story B — Vendor sign-up

As a vendor owner **I want** to register and create a profile with images and pricing **So that** customers can find and contact me.

Acceptance criteria: - Vendor can register with email/phone. - Profile creation wizard collects category, location, description, images, contact details. - Vendor is set to status `pending` until admin verification.

User Story C — Buy a featured listing

As a vendor **I want** a simple way to purchase a featured listing for a week **So that** I appear higher in search results.

Acceptance criteria: - Vendor can purchase boosts in dashboard via payment gateway. - Boosted listings appear in a promoted slot for configured duration.

5. Data Model (Core Entities)

Vendor

- id (uuid)
- name
- slug
- category_id (FK)
- subcategory (string)
- description
- address
- city
- neighborhood
- lat, lng
- phone_number
- email
- website
- social_links {instagram, facebook}
- images []
- tags []
- price_tier (low/medium/high) or estimated_price_range
- rating (float)
- reviews_count (int)
- verified (boolean)
- status (pending/active/suspended)
- subscription_plan_id (FK)
- created_at, updated_at

Category

- id
- name (venue, catering, photographer, DJ, fashion, fabrics, favours, drinks, planners)
- slug

User

- id
- name
- email
- phone
- role (user/vendor/admin)
- password_hash
- favorites []
- created_at

Lead

- id
- vendor_id
- user_id (nullable if guest)
- contact_method (phone/email/in-app)
- message
- status (new/contacted/closed)
- source (search, ad, featured)
- created_at

Boost / Promotion

- id
- vendor_id
- type (featured, banner)
- start_date
- end_date
- transaction_id

Subscription

- id
- vendor_id
- plan (basic/standard/premium)
- price
- renewal_date
- status

Review

- id

- vendor_id
 - user_id
 - rating
 - title
 - body
 - created_at
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6. API (Representative Endpoints)

Auth - POST /auth/register (user/vendor) - POST /auth/login - POST /auth/verify-phone

Search & Listings - GET /listings?category=venue&lat=...&lng=...&q=East%20Legon&price=low - GET /listings/{slug} - POST /listings/{id}/favorite (user)

Vendor Management - POST /vendors (create profile) - GET /vendors/{id}/dashboard (secured) - PUT /vendors/{id} (update)

Leads & Contact - POST /vendors/{id}/lead — {name, phone, email, message, event_date} - GET /vendors/{id}/leads (vendor)

Billing & Promotions - POST /billing/subscribe - POST /billing/purchase-boost - GET /vendors/{id}/billing

Admin - GET /admin/vendors?status=pending - POST /admin/vendors/{id}/approve - GET /admin/analytics

Note: Use JSON for request/response and OAuth2 / JWT for authenticated endpoints.

7. UI / UX (Pages & Components)

Public pages

- Home / Search bar & category tiles
- Category results page (list + map toggle)
- Listing detail page (gallery, description, tags, contact actions, reviews, similar vendors)
- About / How it works / Pricing for vendors

Auth & Vendor pages

- Sign-in / Sign-up
- Vendor onboarding wizard (4 steps: Basic info, Location, Media, Payment & Plan)
- Vendor dashboard (profile editor, leads list, analytics, billing)

Admin pages

- Admin dashboard (pending vendors, financials, reported issues)

- Content moderation & verification

UX Notes - Mobile-first responsive design - Map interaction: cluster pins in dense areas - Quick actions: Call, Message, Request Quote

8. Search & Geo Requirements

- Store `lat/lng` for listings; use geospatial index (Postgres PostGIS or MongoDB geospatial).
 - Support search by text (name, tags) and by proximity radius (e.g., within 5km).
 - Ranking: boosted listings > relevant keywords > proximity > rating.
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9. Monetization Implementation Details

- **Subscriptions:** recurring billing (Stripe, Flutterwave, Pystack). Store subscription state and invoices.
 - **Featured Listings:** time-boxed promotions with tracking of start/end and guaranteed placement.
 - **Per-Lead Billing:** store lead events; flag leads billed; vendor can view billed/unbilled leads.
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10. Non-functional Requirements

- **Performance:** Search responses < 500ms for typical queries; vendor profile < 200ms.
 - **Availability:** 99.5% uptime for MVP.
 - **Scalability:** Design database and API to scale horizontally; caching layer (Redis) for frequent queries.
 - **Security:** HTTPS only, encrypt secrets, secure JWT, rate limiting, input validation, protect against XSS/SQL injection.
 - **Data privacy:** GDPR-like basic privacy, vendor data controls, delete requests support.
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11. Tech Stack Recommendations

- **Frontend:** React (Next.js) or React Native / Expo for mobile. Use Tailwind CSS.
 - **Backend:** Node.js + Express / NestJS or Django REST Framework
 - **DB:** PostgreSQL (+ PostGIS) or MongoDB (with geospatial indexing)
 - **Storage:** S3-compatible (AWS S3, DigitalOcean Spaces) for images
 - **Auth & Payments:** JWT, OAuth optional; Payments via Stripe / Pystack / Flutterwave
 - **Search:** PostgreSQL full-text / Elasticsearch for advanced search (optional later)
 - **Hosting:** Vercel (frontend) + Heroku / Render / AWS EC2 / DigitalOcean (backend)
 - **Queue & Cache:** Redis for caching, background jobs (email, lead distribution)
 - **Analytics:** Google Analytics + Mixpanel / Amplitude
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12. MVP Roadmap & Timeline (Example)

Phase 0 - Setup & Import (1-2 weeks) - Project scaffolding, DB schema, import your vendor list as raw data.

Phase 1 - Core MVP (4-6 weeks) - Search & discover, listing pages, vendor onboarding, lead capture, vendor dashboard, admin panel, basic billing.

Phase 2 - Polish & Monetize (2-4 weeks) - Add subscriptions, boosts, invoices, email flows, analytics.

Phase 3 - Booking & Payments (4-6 weeks) - Add booking calendar, payments, commission flows, chat.

Adjust timelines to team size; an experienced small team (2–4 devs) should hit MVP in ~6–10 weeks.

13. Testing & QA

- Unit tests for core services, integration tests for APIs.
 - End-to-end tests for search and lead flows (Cypress / Playwright).
 - Security audits: dependency checks & OWASP scanning.
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14. Acceptance Criteria (Platform Ready)

- Users can search by category + location and contact vendors.
 - Vendors can register, create profiles, and receive leads.
 - Admin can approve vendors and view platform metrics.
 - Billing for subscriptions & boosts is functional.
 - Initial dataset (your supplied vendors) imported and deduplicated.
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15. Analytics & KPIs to track

- MAU, DAU
 - Number of leads/month
 - Conversion rate (lead → booking)
 - Revenue by source (subscriptions vs boosts vs leads)
 - Avg revenue per vendor
 - Top search terms & categories
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16. Deployment & Monitoring

- CI/CD pipeline (GitHub Actions) to run tests and deploy.

- Monitoring: Sentry for error tracking, Prometheus / Grafana for metrics, uptime monitor.
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17. Next immediate steps (what I can do for you now)

1. Clean & dedupe your vendor list and transform into CSV/JSON seed data.
2. Produce a minimal DB schema SQL + seed script from your dataset.
3. Produce Postman collection / OpenAPI spec for the API endpoints above.
4. Provide a 6-week sprint plan with tasks (if you want to develop now).

Tell me which next step you want and I will generate it.

End of functional specification (v0.1)